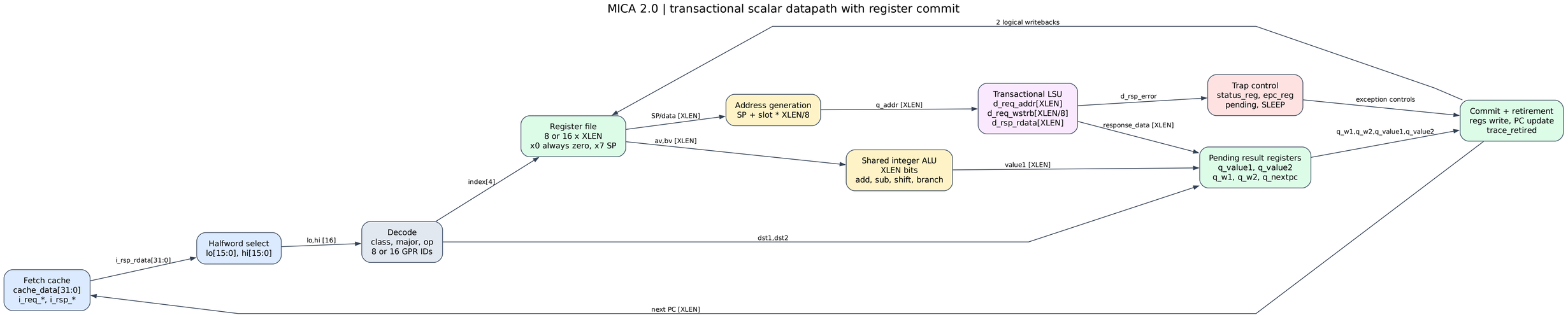
MICA 2.0 | transactional scalar datapath with register commit
2 logical writebacks
Fetch cache
cache_data[31:0]
i_req_*, i_rsp_*
i_rsp_rdata[31:0]
Halfword select
lo[15:0], hi[15:0]
lo,hi [16]
Decode
class, major, op
8 or 16 GPR IDs
index[4]
Register file
8 or 16 x XLEN
x0 always zero, x7 SP
SP/data [XLEN]
av,bv [XLEN]
Address generation
SP + slot * XLEN/8
q_addr [XLEN]
Shared integer ALU
XLEN bits
add, sub, shift, branch
value1 [XLEN]
Transactional LSU
d_req_addr[XLEN]
d_req_wstrb[XLEN/8]
d_rsp_rdata[XLEN]
d_rsp_error
response_data [XLEN]
Trap control
status_reg, epc_reg
pending, SLEEP
exception controls
Pending result registers
q_value1, q_value2
q_w1, q_w2, q_nextpc
q_w1,q_w2,q_value1,q_value2
Commit + retirement
regs write, PC update
trace_retired
dst1,dst2
next PC [XLEN]